

The Dataset grid

The surface where a finance person looks at rows and decides whether the control is right — *“the most important surface in the product.”* It reads, it copies, and it is where a new formula column gets written: as a **Rule**, placed in a step that has a **Trigger**. Everything below is live. Drag a header to reorder, drag its edge to resize, select a range and press ⌘C, or press + at the end of the header row to compose a column.

Control

PEP Settlement & Bank Confirmation

Project

PEP ZA – Settlement

Environment

Dev

Timezone

Africa/Johannesburg

13

Datasets

17

steps

43

rules

DATA

CONTEXT

RULES

D06

Merchant report (normalized)

D06 · Merchant report (normalize)

transactionalnormalizedinternaltransaction

VIEWS

V0 Default 990

V1 RG_U01_PEP 850

Filter rows...

Copy

Fit widths

Reset layout

AS RECEIVEDWRITTEN BY A RULEFROM A SIDEPROPOSEDSYSTEM

84	2026-07-29 11:15:40	GW2607290002	RC26072910001	BT274259200	2726093164	PEP0934	R14535929	7405*****1260	750.84	0.02	14.64	
85	2026-07-29 10:06:15	GW2607290014	RC26072910013	BT338163910	2787671965	PEP0819	R11361999	8612*****1147	350.74	0.02	6.84	
86	2026-07-29 13:29:11	GW2607290024	RC26072910023	BT486059014	2732862276	PEP0287	R95171818	6004*****4225	250.13	0.02	4.88	
87	2026-07-29 11:55:53	GW2607290013	RC26072910012	BT475415326	2712510810	PEP0391	R89198672	4028*****8138	1,200.78	0.02	23.42	
88	2026-07-29 09:30:47	GW2607290010	RC26072910009	BT826613883	2719703720	PEP0287	R36620744	1919*****2176	99.12	0.02	1.93	
89	2026-07-29 19:47:52	GW2607290003	RC26072910002	BT470241762	2784959250	PEP0668	R52994840	8199*****2696	199.29	0.02	3.89	
90	2026-07-29 07:17:30	GW2607290008	RC26072910007	BT481910246	2746691169	PEP0142	R37457561	9955*****7213	1,200.98	0.02	23.42	
91	2026-07-29 20:12:36	GW2607290012	RC26072910011	BT529286619	2790994616	PEP0287	R18015237	8586*****4432	199.56	0.02	3.89	
92	2026-07-29 07:17:03	GW2607290027	RC26072910026	BT156299643	2759925576	PEP1077	R24276413	6852*****2956	99.36	0.02	1.94	
93	2026-07-29 13:49:33	GW2607290018	RC26072910017	BT178332494	2717726784	PEP0287	R20412828	9540*****4130	299.10	0.02	5.83	
94	2026-07-29 11:53:35	GW2607290023	RC26072910022	BT248753333	2782339039	PEP0512	R15353243	8449*****1331	149.44	0.02	2.91	
95	2026-07-29 08:26:30	GW2607290021	RC26072910020	BT694761654	2729926633	PEP0703	R33776277	8012*****6732	149.15	0.02	2.91	
96	2026-07-29 11:54:56	GW2607290019	RC26072910018	BT173529310	2713783716	PEP0455	R63467620	6043*****7686	49.33	0.02	0.96	

104 rows · 30 columns · most recent 120 of 990

⌘C values · Shift⌘C with headers · click a header to select its column

ANATOMY

Column header 46PX

- Provenance rail** — 2px, full width, at the very top. Colour says where the column came from.
- Meta line** — 9.5px mono, uppercase: the column's `type`, then its key badge.
- Name** — 11.5px mono 600. Real column names are `SNAKE_CASE`; never re-case them.
- f** appears when a rule writes the column. Click it for the rule and its trigger.
- Resize handle** — 7px hit area straddling the divider, 2px accent line on hover.

Row 32PX

- Gutter** — 54px, sticky left, row number in 10.5px mono. Click selects the row.
- Money and numerics** right-aligned, mono, `tabular-nums`, thousands separated.
- Status** renders as a pill, coloured by meaning — not by string.
- Empty** is an em-dash at 40% ink, never a blank cell. Blank reads as "not loaded".

Selection

- Click a cell, drag or **Shift-click** to extend. Arrows move, **Shift+arrows** extend.
- Header click selects the column, gutter click the row, corner selects all.
- Wash is `---accent-wash`; the anchor cell carries a 2px inset accent ring.
- The status bar counts the selection and sums it when every cell is numeric.

THE RAILS — WHAT THE COLOURS MEAN

This is the one piece of information design the grid adds beyond a normal spreadsheet, and it exists to carry the product's central claim into the table itself: **what finance sees is what runs**. A person scanning the header row can tell, without clicking anything, which numbers arrived from a counterparty and which ones Simetrik computed.

RAIL	ORIGIN	MEANS
Gray 200	as received	Came in on the file exactly as it is. No rule touched it.
Blue 700	written by a rule	A rule in this Dataset's steps computed it. The f marker opens the rule and the trigger that fires it.
Blue 400	from a side	Inherited from side A or side B of a reconciliation — the <code>a.</code> / <code>b.</code> prefixes.
Violet 700, dashed	proposed	Composed but not run. A new Version of a step that has not executed. Dashed because it is not yet a fact.
Gray 100	system	Provenance and reconciliation bookkeeping. Present on every Dataset, deliberately quiet.

INTERACTIONS

gesture	does	feedback
drag header	Reorders the column	Header drops to 40% opacity, a 2px accent drop line snaps to the nearest boundary, a mono ghost label follows the cursor
drag header edge	Resizes, 56px minimum	Accent divider, <code>col-resize</code> cursor held on the whole component for the length of the drag
double-click edge	Fits that column to its widest cell	Width animates nothing — it snaps. Measured on the canvas, capped at 420px
click header	Selects the whole column	Header and cells wash accent; status bar switches to column stats
⌘C / Ctrl-C	Copies the selection as TSV — pastes straight into Excel or Sheets	Toast naming exactly what went to the clipboard
Shift⌘C	Same, with the column names as a header row	Toast says "with headers"
+ in header row	Opens the composer	Panel slides from the right over the grid, 220ms, respects <code>prefers-reduced-motion</code>
click f	Shows the rule that writes the column and the trigger that fires it	Popover anchored under the marker; dismisses on outside click or Esc

TOKENS

The ground is paper, in both themes. A spreadsheet is a document surface, and the people who live in this screen already read white all day in Excel. So the neutrals are the brandbook's true greys with no hue bias, and the dark theme is a neutral charcoal rather than a tinted one. Blue is spent only where the product is *saying* something — a selection, a rule, a focused control. It is never the ground.

It opens **Light** regardless of the operating system, because that is the surface this component is imitating. The switch at the top of the page overrides it and the choice sticks.

token	light	dark	used for
--surface	 #FFFFFF	 #1F1F1F	Cell ground. Neutral in both themes
--surface-2	 #F8F8F8	 #292929	Header, gutter, toolbar, status bar
--surface-3	 #F3F3F3	 #333333	Inputs, chips, hover fills
--line	 #DEDEDE	 #3A3A3A	Cell gridlines
--line-strong	 #C4C4C4	 #4D4D4D	Component border, gutter divider, header underline
--ink	 #111111	 #EDEDED	Cell values
--ink-2	 #5E5E5E	 #ABABAB	Secondary, status bar
--ink-3	 #919191	 #7A7A7A	Column meta, empty cells
--accent	 #3838F9	 #5968F9	Selection, focus, active view, the rule marker
--accent-wash	 #F3F7FF	 #2C3050	Selected cells
--ok	 #0C9879	 #11D9AD	Reconciled, settled, resolved
--brk	 #D42C3C	 #FF6472	Unreconciled, a break
--attention	 #B26808	 #FFB454	Open anomaly, ageing
--proposed	 #BA0BFF	 #C83CFF	A column composed but not yet run
--row-h	32px	32px	Row height
--head-h	46px	46px	Header height
--gutter-w	54px	54px	Row-number gutter
--cell-pad	10px	10px	Horizontal cell padding

WHAT THIS COMPONENT FORCES A DECISION ON

Can a column be authored from the grid at all? BLOCKING

F3 §11 reads **no** for v1: the chat is the only way logic changes, because hand-edits create two sources of truth. But a spreadsheet with a + on the header row is the single most expected affordance in this surface, and a finance person adding a column is describing intent, not writing code.

What is built here is the compromise: the grid **composes** and the result is a **proposal** — a violet dashed column and a banner saying which step it changed and that the step has not run. Nothing is committed from the grid. If the team lands on chat-only, the + stays and hands the composed rule to the chat instead of applying it.

A Rule has no trigger — a step does MODEL

`objects/ruleset.md` §3 puts the trigger on the step, not the rule, and a step with zero

triggers is a validation error rather than a configuration. So "a formula column with its trigger" cannot be one field. The composer asks it as two: **append to a step that already runs** — and inherit its triggers, shown in full so nothing is implicit — or **start a new step**, which then must pick at least one trigger from the closed enum of six.

Appending is the default and it is the right one. The overwhelming majority of new columns belong to the step that already builds the Dataset, and forcing a trigger choice every time would teach people to pick one without reading it.

Are column order and width per person, or per Dataset? NEEDS DECIDING

Reordering and resizing feel like view state, and treating them as personal is the cheap answer. But an operator who arranges a break-investigation layout wants their team to open the same one, and a Dataset with 33 columns has no useful default order. The likely shape is a saved arrangement per **View**, owned like the View is — which makes column order part of the control's definition, and therefore versioned.

How large is too large to show? NEEDS DECIDING

`objects/dataset.md` §10 raises it and this component does not answer it. It renders every row it is handed. The demo bundle caps each Dataset at 120 rows; production Datasets run to hundreds of millions. Windowed rendering is a build concern, but **what the status bar says when the grid is showing a slice of a much larger truth** is a design one, and it is unresolved.

Can a cell be edited directly? OUT OF SCOPE, DELIBERATELY

Not built. `01-information-architecture.md` §7 has it open and it is a bigger question than it looks: a direct edit is a data mutation outside the chat, needing its own attribution and approval. The one legitimate route already exists in the model — an `input` rule, which declares that a person's judgement goes in that column. The composer offers `input` as a rule type for exactly that reason.

WHERE THE DATA COMES FROM

Everything in the grid is the **PEP Settlement & Bank Confirmation** control from `09-demo-runtime-spec-PEP.md`, generated by running each Ruleset's rules in its stated arrangement — so every derived column holds what that rule would actually have written. All thirteen Datasets are in the picker. The reconciliation cascade fires on all three of its passes, Youth Day pushes a settlement's expected bank date from 16 to 17 June through the calendar lookups, and one day in D12 breaks by more than the 50-unit tolerance.